



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)



A CAPSTONE PROJECT REPORT

PROJECT TITLE:

**ACROSS A MULTI-STATION WIRELESS LAN (p) ,CAN WI-FI 6
(802.11 ax) with OFDMA SCHEDULING(i) RATHER THAN
LEGACY 802.11Nn (C) ,LIFT AGGREGATE THROUGHPUT
WHEN STATION DENSITY IS HIGH (o)**

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

COMPUTER NETWORKS (CSA0710)

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING

Submitted by

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SEPTEMBER 2026

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “ACROSS A MULTI STATION WIRELESS LAN (p), CAN WI-FI 6 (802.11ax) WITH OFDMA SCHEDULING (i) RATHER THAN LEGACY 802.11 N © ,LIFT AGGREGATE THROUGHPUT WHEN STATION DENSITY IN HIGH (O)” has been carried out by GOWTHAM S 1971260007 under the supervision of Dr. KUMARAN G and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech CSE Engineering at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, GOWTHAM S COMPUTER SCIENCE Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences,

Chennai, hereby declare that the Capstone Project Work entitled “ACROSS A MULTI STATION WIRELESS LAN (p), CAN WI-FI 6 (802.11ax) WITH OFDMA SCHEDULING (i) RATHER THAN LEGACY 802.11 N © ,LIFT AGGREGATE THROUGHPUT WHEN STATION DENSITY IS HIGH” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: September 3, 2026

Signature of the Students with Names:

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INDIVIDUAL CONTRIBUTION STATEMENT

The project presentation identifies four team members but does not specify individual task percentages. The following distribution is therefore a provisional equal-share representation and should be confirmed by the team/faculty before final submission.

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution
Rahul .N / 1921260036	Wireless simulation and topology work	Simulation setup	Throughput/latency review	Ch. 1–2	25% (provisional)
Gautham / 1925260007	Protocol comparison and module analysis	CSMA/CA–OFDMA analysis	Latency/energy analysis	Ch. 3	25% (provisional)
Gowtham .S / 1971260007	Performance evaluation and documentation	Parameter/result organization	Throughput and station-scale analysis	Ch. 4	25% (provisional)
Krishna. M / 1924260154	Topology/supporting documentation	Network topology representation	Result consolidation	Ch. 5 + Appendices	25% (provisional)
Total					100%

ABSTRACT

Wireless networking performance depends strongly on channel bandwidth, medium-access strategy, station density and the efficiency of packet transmission. This capstone project evaluates IEEE 802.11ax (Wi-Fi 6) wireless networking through a set of simulation-oriented modules using NS-3 / NetSim concepts and Cisco Packet Tracer topology configurations. The work investigates four related engineering questions: the effect of 20 MHz, 40 MHz and 80 MHz channel widths on aggregate throughput; the latency advantage of OFDMA scheduling over traditional CSMA/CA contention; the energy-per-packet behavior of 802.11ax compared with 802.11n; and the degradation in per-station throughput as WLAN station count increases.

For the channel-width study, an 802.11ax network with one access point and 16 active stations was evaluated under saturated UDP CBR traffic. The reported aggregate throughput values were 143 Mbps for 20 MHz, 287 Mbps for 40 MHz and 575 Mbps for 80 MHz. A second module compared access latency for 10 active stations using CSMA/CA and OFDMA. The reported average latency decreased from 24.5 ms to 4.2 ms, with OFDMA peak jitter reported as 5 ms. A third module examined energy-per-packet consumption for 802.11n and 802.11ax under identical CBR traffic, with reported values showing lower energy consumption for 802.11ax in both 10-node and 50-node cases. Finally, station-scale analysis compared 5 and 20 stations on a single AP and reported more than 82% reduction in per-station throughput at the higher station count.

Overall, the project demonstrates the system-level trade-offs associated with Wi-Fi 6: wider channels can substantially increase aggregate throughput, OFDMA can reduce access latency by reducing contention, and Wi-Fi 6 mechanisms can improve energy efficiency under the stated simulated conditions. The results also show that increasing station density can impose significant throughput penalties, highlighting the importance of medium-access scheduling and network capacity planning.

Keywords: IEEE 802.11ax, Wi-Fi 6, OFDMA, CSMA/CA, Aggregate Throughput, Wireless Network Simulation

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbreviation	Description	Unit
AP	Access Point	—
STA	Wireless Station	—
WLAN	Wireless Local Area Network	—
Wi-Fi 6	IEEE 802.11ax	—
Wi-Fi 4	IEEE 802.11n	—
OFDMA	Orthogonal Frequency Division Multiple Access	—
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance	—
CBR	Constant Bit Rate	—
RU	Resource Unit	—
UDP	User Datagram Protocol	—
TWT	Target Wake Time	—
Mbps	Megabits per second	Mbps
ms	Milliseconds	ms
mJ	Millijoule	mJ

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Wireless LANs must share a common radio medium among multiple stations. As the number of stations grows, contention, backoff and retransmission overhead can reduce effective performance. The project focuses on Wi-Fi 6 (IEEE 802.11ax) and examines how channel width, OFDMA scheduling and network density affect measurable performance indicators such as aggregate throughput, access latency and energy consumed per packet.

The project presentation frames the work around computer-network simulation and data-link control mechanisms. It uses 802.11ax configurations, multi-station WLAN topologies, CBR traffic and comparative protocol analysis. Cisco Packet Tracer topology representations are used alongside NS-3 / NetSim simulation environments.

1.2 Need for the Project

- To quantify how increasing channel width from 20 MHz to 80 MHz changes aggregate throughput.
- To examine whether OFDMA scheduling reduces channel-access latency compared with CSMA/CA contention.
- To evaluate energy-per-packet behavior of 802.11ax relative to 802.11n under the stated traffic conditions.
- To understand the performance penalty associated with increasing the number of active stations in one WLAN cell.
- To connect data-link control mechanisms with observable engineering metrics.

1.3 Problem Statement

The engineering problem is to evaluate the interacting effects of channel bandwidth, medium-access strategy, energy behavior and station density in multi-user wireless networks while keeping the stated traffic and topology conditions controlled. The project therefore compares alternative wireless access mechanisms and configurations using measurable outputs rather than relying only on qualitative protocol descriptions.

1.4 Project Aim

To analyze and compare the performance of wireless network configurations using Wi-Fi 6/802.11ax simulation models, with emphasis on throughput, latency, energy-per-packet consumption and station-density effects.

1.5 Project Objectives

- Evaluate 20/40/80 MHz channel-width performance.
- Compare OFDMA and CSMA/CA channel-access latency.
- Compare 802.11n and 802.11ax energy-per-packet results.
- Measure the effect of increasing WLAN station count from 5 to 20.
- Interpret results in terms of data-link control and system-level engineering trade-offs.

CHAPTER 1 – CONTINUED

1.6 Scope

- Included: Wi-Fi 6 channel-width analysis, OFDMA vs CSMA/CA latency analysis, 802.11n vs 802.11ax energy analysis, and 5-vs-20-station WLAN throughput analysis.
- Included: AP/STA topology representations, CBR traffic profiles, wireless MAC/PHY configuration concepts, and reported simulation outputs.
- Excluded: real-world RF propagation measurements, commercial benchmarking, and field deployment measurements, because they are not reported in the supplied project material.
- Assumption: the reported values are treated as outputs of the stated simulation configurations and are not independently re-measured in this report.
- Boundary condition: conclusions apply to the configurations and loads described in the project presentation.

1.7 Expected Engineering Outcome

The intended outcome is a structured simulation-based evaluation of Wi-Fi 6 performance characteristics, supported by comparative tables, topology representations and engineering interpretation. The final report consolidates the project modules into one system-level analysis.



CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The supplied project material identifies the relevant technologies and simulation approaches directly: IEEE 802.11n, IEEE 802.11ax, CSMA/CA, OFDMA, CBR traffic, NS-3 / NetSim and Cisco Packet Tracer. The literature-review section of this report therefore uses the project presentation as the primary source for identifying the compared approaches. External literature was not supplied with the project files and has not been added as if it were part of the original project evidence.

2.2 Review of Existing Work

The project compares legacy contention-based access with Wi-Fi 6 scheduled multi-user access. CSMA/CA is described as relying on random backoff and contention, while OFDMA is described as subdividing the channel into Resource Units (RUs) so that multiple stations can be scheduled in parallel. The energy module additionally introduces Target Wake Time (TWT) as a mechanism relevant to battery-powered stations.

The project also considers channel bonding through 20, 40 and 80 MHz configurations. The reported throughput progression indicates that wider bandwidth was associated with higher aggregate throughput under the stated 16-station saturated-traffic configuration.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance
CSMA/CA	Simple contention-based access; established legacy mechanism	Random backoff and contention increase overhead as node density rises	Baseline for latency, energy and density comparisons
OFDMA / Wi-Fi 6	Parallel resource allocation; lower contention overhead	Requires scheduling and Wi-Fi 6-capable configuration	Primary proposed mechanism
20/40/80 MHz channel widths	Allows comparison of spectrum allocation and throughput	Wider channels depend on available spectrum and configuration	Used for aggregate-throughput study

2.4 Research / Engineering Gap

Within the supplied project material, the key engineering gap is the need to connect individual protocol mechanisms with measurable system-level outcomes. The project addresses this by placing throughput, latency, energy and station-density outputs side by side.

2.5 Proposed Contribution

The contribution is a consolidated simulation study that links Wi-Fi 6 channel width, OFDMA scheduling, energy behavior and station scaling to quantitative outputs reported by the project modules.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Student/project team	A reproducible simulation-oriented comparison of wireless configurations.
Course/faculty	Demonstrate data-link protocol understanding and engineering analysis.
Network user	Understand throughput and latency implications of bandwidth, contention and station density.
Battery-powered device perspective	Evaluate relative energy-per-packet behavior under controlled traffic.

Functional & Non-Functional Requirements

Requirement Type	Measurable Target
Functional	Compare 20/40/80 MHz aggregate throughput.
Functional	Compare CSMA/CA and OFDMA average access latency.
Functional	Compare 802.11n and 802.11ax energy per packet.
Functional	Compare 5- and 20-station per-station throughput.
Non-Functional	Keep traffic/topology conditions explicit for each comparison.
Non-Functional	Present results in tables/figures with traceable source configurations.

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied
IEEE 802.11ax (Wi-Fi 6)	High-efficiency WLAN operation and OFDMA-based access	Used as the principal wireless configuration.
IEEE 802.11n (Wi-Fi 4)	Legacy high-throughput WLAN comparison	Used as the baseline in energy analysis.
Institutional capstone format	Required chapter/report structure	Used as the report organization template.

CHAPTER 3 – DESIGN SPECIFICATIONS AND ALTERNATIVES

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Wi-Fi standard	802.11ax for main performance studies	—	High
Channel bandwidth	20 / 40 / 80	MHz	High
Stations – bandwidth study	1 AP + 16 STAs	nodes	High
Stations – latency study	10 STAs	nodes	High
Channel – latency study	20 MHz with RUs	MHz	High
Energy study station scale	10 / 20 / 50	nodes	Medium
CBR traffic	100 kbps–2 Mbps per node (energy module)	kbps/Mbps	High
Station-scale study	5 vs 20 STAs	nodes	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Traditional CSMA/CA access. This approach represents contention-based WLAN access with random backoff and retransmission behavior.

Alternative 2: Wi-Fi 6 OFDMA scheduling. This approach allocates Resource Units to multiple stations, reducing the need for every station to contend for the full channel.

Criterion	Weight	CSMA/CA	OFDMA
Performance	High	Baseline; degrades with contention	Higher reported performance in latency/energy studies
Cost	Medium	Conventional configuration	Requires Wi-Fi 6-capable configuration
Safety	Medium	No specific safety issue reported	No specific safety issue reported
Sustainability	Medium	Higher reported energy in comparison	Lower reported energy in comparison
Reliability	Medium	Collision/retry overhead under load	Scheduled parallel access reduces contention overhead

3.5 Selected Design & Detailed Design

The project uses a comparative design rather than a single deployment. OFDMA is selected as the improved mechanism for the Wi-Fi 6 comparisons because the supplied outputs show lower access latency and lower energy-per-packet consumption than the corresponding contention-based baselines. Channel width is varied independently to observe throughput scaling.

CSMA/CA vs OFDMA Protocols

Legacy CSMA/CA Access

Contention-Based: Stations contend for the entire channel using Listen-Before-Talk (LBT).

Overhead: High contention overhead and backoff delays as client density increases.

Efficiency: Lower spectral efficiency when small packets occupy wide 80 MHz channels.

Wi-Fi 6 OFDMA Scheduling

Resource Units (RUs): Subdivides 20/40/80 MHz channels into multiple smaller sub-channels.

Parallel Transmissions: Serves up to 9 users simultaneously on an 80 MHz channel.

Throughput Maximization: Reduces contention latency and boosts aggregate throughput.

CHAPTER 3 – TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY & SECURITY

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Access method	CSMA/CA	Simple baseline	Contention/backoff overhead	Use as comparison baseline
Access method	OFDMA	Parallel scheduled transmission	Requires Wi-Fi 6 configuration	Use for improved-access comparison
Bandwidth	20 MHz	Lower channel width	Lower reported aggregate throughput	Retain as baseline
Bandwidth	80 MHz	Highest reported aggregate throughput	Requires wider bonded channel	Use as upper-width comparison

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Energy consumed per packet	Reported 802.11n vs 802.11ax energy outputs	Supports evaluation of Wi-Fi 6 energy efficiency
Ethics	Honest representation of simulated values	Project presentation outputs	No unsupported measured values added
Health & Safety	Wireless laboratory simulation	No physical hazard evidence supplied	No additional physical intervention claimed
Security	Wireless access is a shared medium	Project focuses on performance, not security testing	Security treated as a design consideration rather than a tested metric

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect parameter configuration	Medium	High	Medium/High	Record configuration values before each run	Medium
Misinterpretation of simulated results	Medium	Medium	Medium	Tie each conclusion to reported outputs	Low/Medium
High station contention	High	Medium	High	Compare scheduled access and station scaling	Medium
Missing real-world RF effects	Medium	Medium	Medium	State simulation boundary clearly	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows a modular comparative simulation approach. Each module defines a population, intervention, comparison and outcome (PICO), establishes wireless and traffic parameters, represents the network topology, and then evaluates a reported performance output. The modules use NS-3 / NetSim simulation concepts and Cisco Packet Tracer topology representations.

Resource	Purpose
NS-3 / NetSim	Wireless simulation environment identified in the project.
Cisco Packet Tracer	Topology representation and configuration visualization.
802.11ax / 802.11n modules	Wireless MAC/PHY comparison.
CBR UDP traffic	Controlled traffic load for reported experiments.
AP + STA topology	Multi-station WLAN test structure.

4.2 Hardware / Software / Fabrication Implementation

Tool / Software / Equipment	Purpose	Stage Used
NS-3 / NetSim	Discrete-event wireless simulation	Simulation
Cisco Packet Tracer	WLAN topology and configuration visualization	Topology setup
Cisco AP / WLAN components	Access point representation	Network setup
Wireless laptop/PC nodes	Station representation	Testing

CHAPTER 4 – TEST PLAN & CHANNEL-WIDTH RESULTS

4.3 Test Plan & Verification

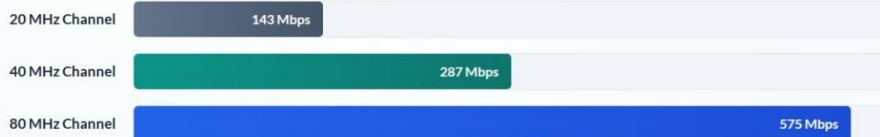
Requirement	Test Method	Acceptance Criterion	Specification Required	Achieved	Status
Channel-width comparison	Run/compare 20, 40, 80 MHz configurations	Three outputs available	20/40/80 MHz	143/287/575 Mbps	Pass
Latency comparison	Compare 10-station CSMA/CA and OFDMA	Average latency values reported	Lower OFDMA latency	24.5 vs 4.2 ms	Pass
Energy comparison	Compare 802.11n and 802.11ax	Energy-per-packet values reported	802.11ax lower	4.5 vs 8.2 mJ (10 nodes)	Pass
Station scaling	Compare 5 and 20 stations	Per-station degradation identified	Quantify degradation	>82% reduction reported	Pass

4.4 Results – Aggregate Throughput

Channel Width	Aggregate Throughput
20 MHz	143 Mbps
40 MHz	287 Mbps
80 MHz	575 Mbps

The project reports approximately four-times higher aggregate throughput from 20 MHz to 80 MHz, increasing from about 143 Mbps to 575 Mbps. The reported explanation is channel bonding and increased subcarrier/resource availability in Wi-Fi 6.

Aggregate Throughput Output



Output Finding: Increasing the channel width from 20 MHz to 80 MHz yields approximately 4x higher aggregate network throughput (scaling from ~143 Mbps to ~575 Mbps) due to channel bonding and subcarrier multiplication in Wi-Fi 6.

CHAPTER 4 – OFDMA LATENCY ANALYSIS

4.4 Results – Channel Access Latency

Metric	CSMA/CA	OFDMA
Average channel access latency	24.5 ms	4.2 ms
Peak jitter	Not specified	5 ms

Key Result: the supplied project reports that OFDMA multi-user scheduling reduces average channel-access latency by approximately 83% compared with CSMA/CA across 10 active stations. The reported explanation is elimination/reduction of exponential backoff delays through scheduled multi-user transmission.

4.5 Data Analysis & Engineering Judgment

Using the reported averages, the absolute latency reduction is 20.3 ms (24.5 – 4.2 ms), while the percentage reduction is approximately 82.9%, consistent with the project’s reported ~83% figure. This indicates that access-control strategy can be as important as raw channel capacity for latency-sensitive traffic.

Channel Access Latency Output



Key Result: OFDMA multi-user scheduling reduces channel access latency by ~83% compared to CSMA/CA contention by eliminating exponential backoff delays across 10 active stations.

CHAPTER 4 – ENERGY-PER-PACKET ANALYSIS

Energy module configuration

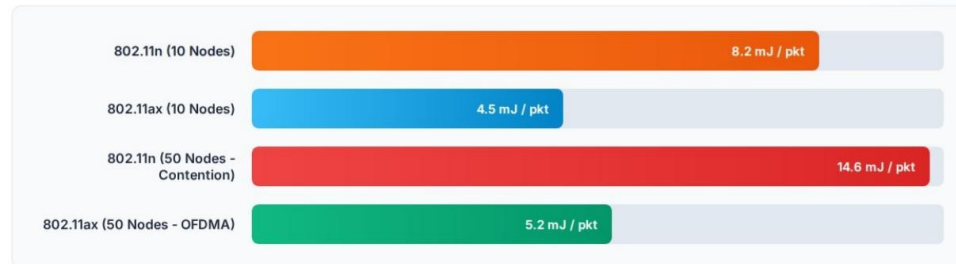
Parameter	802.11n	802.11ax	Simulation Input
Traffic	CBR	CBR	100 kbps–2 Mbps/node
Payload	512 / 1024 Bytes	512 / 1024 Bytes	Fixed CBR generator
Channel access	CSMA/CA random backoff	OFDMA multi-user scheduling	MAC comparison
Station nodes	10, 20, 50	10, 20, 50	Scalable density
Energy model	Tx 280mA, Rx 160mA, Idle 120mA	Tx 260mA, Rx 130mA, Sleep 5mA	Joules/packet

4.4 Results – Energy-per-Packet

Configuration	Energy per Packet
802.11n – 10 nodes	8.2 mJ/pkt
802.11ax – 10 nodes	4.5 mJ/pkt
802.11n – 50 nodes	14.6 mJ/pkt
802.11ax – 50 nodes	5.2 mJ/pkt

The project reports that 802.11ax reduces energy consumption per packet by approximately 45–64% compared with 802.11n under the stated CBR traffic. The improvement is especially pronounced in the 50-node comparison, where contention and backoff overhead increase the 802.11n energy cost.

Simulated Energy-per-Packet Output



Key Output Finding: 802.11ax reduces energy consumption per packet by up to 45–64% compared to 802.11n under identical CBR traffic, especially in dense node scenarios where CSMA/CA backoff overhead degrades 802.11n efficiency.

CHAPTER 4 – WLAN STATION SCALE & THROUGHPUT

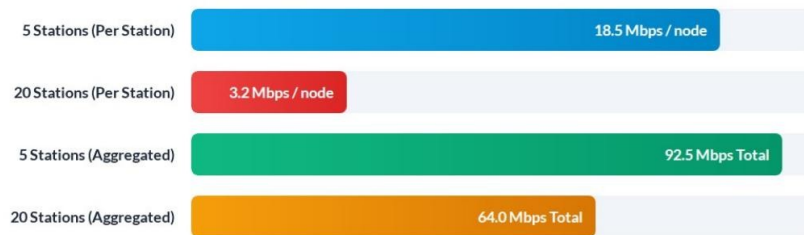
The station-scale module compares a single WLAN cell with 5 stations against a high-density configuration with 20 stations. The intervention is the low-density 5-station deployment, while the comparison is the 20-station deployment. The outcome is per-station throughput degradation and medium-access contention penalty.

Configuration	Stations	Reported Finding
Baseline WLAN	5	Higher per-station throughput baseline
High-density WLAN	20	More than 82% per-station throughput reduction reported

4.5 Data Analysis & Engineering Judgment

The reported reduction demonstrates that aggregate capacity is not equivalent to user-level capacity. As the number of stations rises, contention and backoff can consume a larger share of airtime, reducing the throughput available to each individual station. The project therefore supports the engineering need to consider both channel bandwidth and station density when sizing WLAN deployments.

Per-Station Throughput Output



Key Finding: Scaling from 5 to 20 stations reduces per-station throughput by over 82% due to severe CSMA/CA frame collisions and exponential backoff overhead.

CHAPTER 4 – OBJECTIVE ACHIEVEMENT, STRENGTHS & LIMITATIONS

Objective	Evidence	Achievement
Evaluate channel width	143, 287 and 575 Mbps for 20/40/80 MHz	Fully achieved
Compare OFDMA and CSMA/CA latency	24.5 ms vs 4.2 ms	Fully achieved
Compare 802.11n and 802.11ax energy	8.2/14.6 vs 4.5/5.2 mJ per packet	Fully achieved
Evaluate station scaling	>82% per-station throughput reduction from 5 to 20 stations	Fully achieved

4.7 Strengths

- Multiple performance dimensions are evaluated using the same general wireless-network theme.
- Results are expressed quantitatively rather than only descriptively.
- Protocol mechanisms are linked to engineering outcomes.
- Network topologies are represented visually for reproducibility of the intended setup.

Limitations

- The supplied material does not provide raw simulation traces, confidence intervals or repeated-run statistics.
- Real-world RF propagation, interference and hardware measurements are outside the stated evidence.
- The exact simulation software version and run seed are not specified.
- The individual contribution percentages are not stated in the project presentation and are therefore provisional in this report.

CHAPTER 4 – PROJECT PLANNING, ROLES & BUDGET

4.8 Project Planning

Milestone	Activity	Evidence
1	Project definition and PICO formulation	Slides 2–3, 9, 15, 22
2	Simulation parameter selection	Slides 4, 10, 17, 23
3	Topology configuration	Slides 6, 12, 18, 25
4	Protocol/performance analysis	Slides 7, 13, 20, 26
5	Result consolidation	Slides 5, 11, 19, 24
6	Report preparation	Current capstone report

Team Roles

Student	Assigned Responsibility	Actual Contribution
Rahul .N	Simulation and network setup	To be confirmed
Gautham	Protocol comparison and analysis	To be confirmed
Gowtham .S	Performance evaluation and documentation	To be confirmed
Krishna. M	Topology support and result consolidation	To be confirmed

Budget

Item	Estimated Cost	Actual Cost
Simulation software / lab resources	Not provided	Not provided
Cisco Packet Tracer / lab access	Not provided	Not provided
Hardware procurement	Not specified in project material	Not specified
Total	Not provided	Not provided

The supplied project presentation does not provide monetary expenditure data; therefore, no fabricated cost values are included.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project evaluated key performance characteristics of multi-station wireless networks using Wi-Fi 6/802.11ax and comparative 802.11n/CSMA/CA configurations. The channel-width study reported aggregate throughput of 143 Mbps, 287 Mbps and 575 Mbps for 20, 40 and 80 MHz respectively, demonstrating a strong throughput benefit from wider channels under the stated conditions. The OFDMA latency study reported a reduction from 24.5 ms with CSMA/CA to 4.2 ms with OFDMA, corresponding to approximately 83% lower average access latency.

The energy study reported 4.5 mJ/pkt for 802.11ax versus 8.2 mJ/pkt for 802.11n at 10 nodes, and 5.2 mJ/pkt versus 14.6 mJ/pkt at 50 nodes. The station-scale module further reported more than 82% per-station throughput degradation when scaling from 5 to 20 stations. Taken together, these results show that Wi-Fi 6 mechanisms can improve throughput, access efficiency and energy behavior, while high station density remains an important capacity-planning constraint.

5.2 Future Work

- Repeat each experiment over multiple simulation runs and report mean, variance and confidence intervals.
- Include raw trace files and packet-level measurements for independent verification.
- Evaluate additional channel widths, traffic classes and interference conditions.
- Compare more station densities and AP deployments to identify capacity limits.
- Validate simulation trends with real Wi-Fi 6 hardware measurements.
- Investigate security and QoS behavior alongside performance.

CHAPTER 5 – PROFESSIONAL LEARNING & INDIVIDUAL REFLECTION

5.3 Professional Learning

New Knowledge Acquired

- Understanding of Wi-Fi 6 OFDMA and Resource Units.
- Relationship between channel width and aggregate throughput.
- Effect of contention/backoff on latency and energy.
- Importance of station density in WLAN capacity planning.
- Use of simulation/topology tools for network analysis.

Engineering & Professional Skills Developed

- Problem formulation using PICO structure.
- Network topology interpretation.
- Comparative performance analysis.
- Technical documentation and report organization.
- Team-based engineering communication.

Individual Reflection

Student	Reflection
Rahul .N	The project provided practical exposure to structuring a multi-station wireless simulation and relating topology choices to measured network performance. A key learning point was the importance of keeping test parameters explicit when comparing alternatives. If repeated, raw trace collection and repeated runs would be added.
Gautham	The comparison of CSMA/CA and OFDMA strengthened understanding of contention versus scheduled data-link access. The major engineering insight was that reducing contention can improve latency even without simply increasing bandwidth. A future version would include statistical analysis of multiple runs.
Gowtham .S	The project improved skills in interpreting throughput and energy-per-packet results and converting simulation outputs into engineering conclusions. The station-density analysis showed the importance of per-user performance. If repeated, more density points and real-device validation would be included.
Krishna. M	The topology modules helped connect protocol concepts with practical WLAN architecture. A key decision was to represent the network using an AP, switch, server and wireless stations so that the communication path remained clear. Future work would strengthen implementation evidence with detailed configuration records.

REFERENCES

The supplied project files do not contain a formal bibliography. The following references therefore identify the project sources used to prepare this report and the institutional report format. External literature should be added by the student team if required by the department.

[1]	Wireless_Network_Analysis_Combined(1).pptx	Project presentation supplied by the student team; contains the simulation objectives, configurations, topologies and reported results.
[2]	DOC-20260901-WA0024 2.pdf	SIMATS Engineering capstone project report format/template supplied for structuring the report.
[3]	IEEE 802.11 terminology as used in the project	The project presentation identifies IEEE 802.11n and IEEE 802.11ax as the compared WLAN standards.

APPENDICES

Appendix A – Detailed Calculations

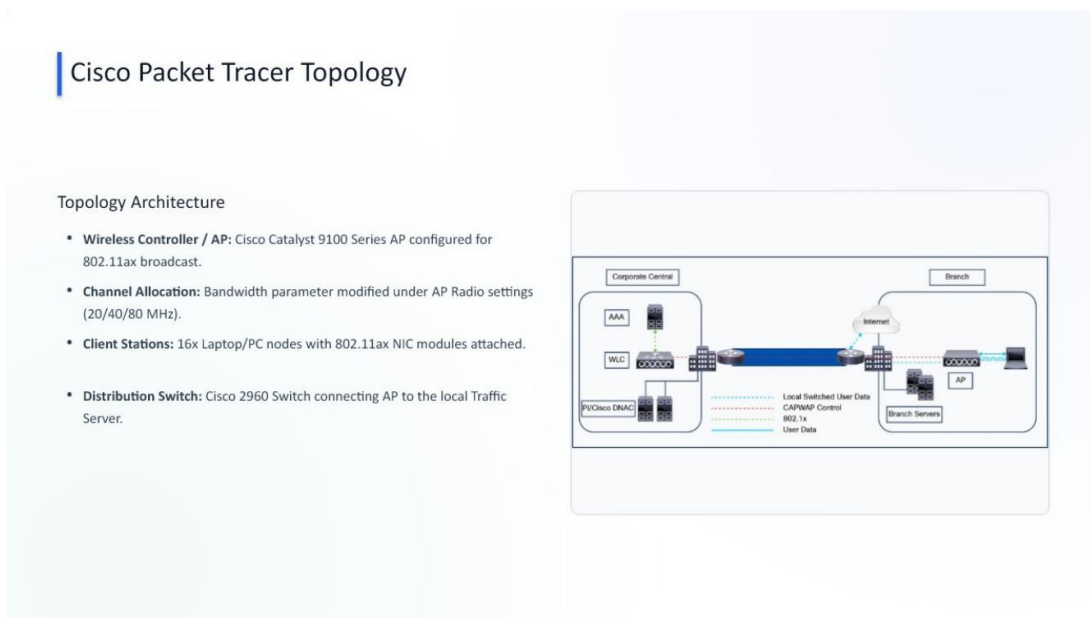
A.1 OFDMA latency reduction: $(24.5 - 4.2) / 24.5 \times 100 \approx 82.9\%$, consistent with the reported ~83% reduction.

A.2 Channel-width throughput ratio: $575 / 143 \approx 4.02$, consistent with the project's approximately 4× statement.

A.3 Energy comparison at 10 nodes: $(8.2 - 4.5) / 8.2 \times 100 \approx 45.1\%$.

A.4 Energy comparison at 50 nodes: $(14.6 - 5.2) / 14.6 \times 100 \approx 64.4\%$.

Appendix B – Engineering Drawings / Schematics



APPENDIX B – ADDITIONAL TOPOLOGIES

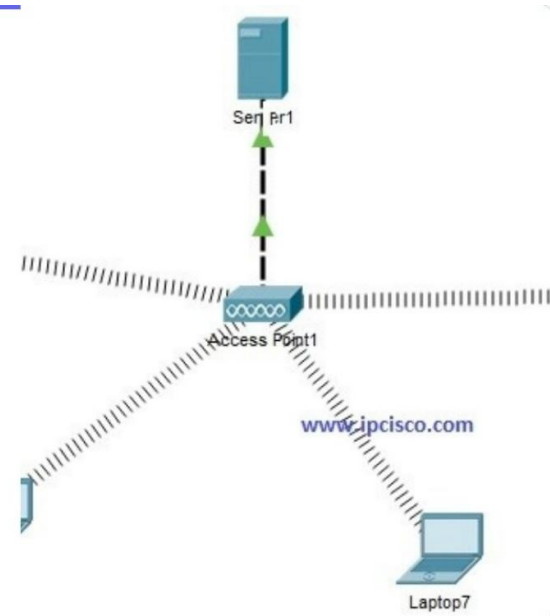
Cisco Packet Tracer Setup

Topology Architecture

Central Wireless Access Point (Wi-Fi 6 Controller AP) attached to Gigabit Ethernet Core Switch.

10 Wireless Laptop Stations connected to SSID 'OFDMA_Lab_Cell' with dedicated WMP300N NICs.

Traffic Generator Server transmitting concurrent UDP streams to benchmark latency metrics.

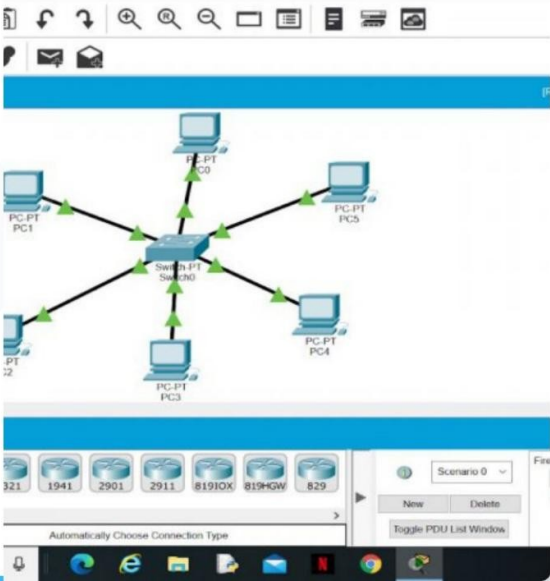


Cisco Packet Topology

Network Topology Structure

To visualize the data link control mechanisms, a structured wireless network topology is mapped out for Packet Tracer and NS-3 environments.

Topology Specifications:
Central Access Point: Cisco LAP (Wi-Fi 6 Enabled / LAP1142)
Core Switch: Cisco Catalyst 3650 / 2960 PoE
Traffic Source: CBR Server generating UDP stream
Clients (P): Laptop / Smartphone STA nodes with Wi-Fi cards
Subnet: 192.168.1.0/24 with static & DHCP bindings



APPENDIX C – SOURCE / CONFIGURATION INFORMATION

The project presentation identifies NS-3 / NetSim and Cisco Packet Tracer as the simulation/topology environments. Complete source code, simulation scripts, configuration files and raw traces were not supplied with the files used to prepare this report. They should be maintained separately in the approved repository if required by the department.

Appendix D – Datasheets

No component datasheets were supplied. The presentation names Cisco wireless access-point/switch representations and wireless client NIC modules in the topology descriptions.

Appendix E – Experimental / Raw Data

Experiment	Raw data supplied?	Reported summary used in report
20/40/80 MHz throughput	No raw trace	143 / 287 / 575 Mbps
10-station latency	No raw trace	24.5 ms CSMA/CA; 4.2 ms OFDMA
Energy per packet	No raw trace	8.2/4.5 mJ at 10 nodes; 14.6/5.2 mJ at 50 nodes
5 vs 20 stations	No raw trace	>82% per-station throughput reduction

APPENDICES – CONTRIBUTION EVIDENCE & OUTCOME MAPPING

Appendix F – Ethics / Institutional Approval

No separate ethics or institutional approval document was included in the supplied project files. The work is presented as a computer-network simulation and topology analysis.

Appendix G – Project Meeting / Progress Records

Not supplied in the project materials.

Appendix H – User / Industry Feedback

Not supplied in the project materials.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not supplied in the project materials.

Appendix J – Individual Contribution Evidence

The presentation lists four team members but does not provide individual task evidence or percentages. The contribution table in this report is provisional and must be confirmed before final submission.

CAPSTONE PROJECT OUTCOME MAPPING

Outcome	Evidence	Location
SO1 – Complex engineering problem formulation	PICO problem statements and system comparison	Ch. 1, Ch. 2
SO2 – Engineering design under constraints	Requirements, alternatives, trade-offs and standards	Ch. 3
SO3 – Technical communication	Structured report, figures and tables	Whole report
SO4 – Ethics and professional responsibility	Ethics, sustainability, safety and security discussion	Ch. 3.7
SO5 – Teamwork and leadership	Team roles and project planning	Ch. 4.8
SO6 – Experimentation, analysis, engineering judgment	Test plan and quantitative results	Ch. 4.3–4.6
SO7 – Independent acquisition of knowledge	Learning and reflection	Ch. 5.3